

Airbnb Price Prediction and Neighborhood Analysis — NYC

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1 Introduction

This project analyzes New York City Airbnb listing data and builds a machine learning model to predict listing prices. The goal is to understand how factors such as location, room type, availability, and review activity influence Airbnb prices.

The project also includes a Tableau dashboard to visualize borough-level and neighborhood-level pricing trends.

2 Dataset

The dataset used is the New York City Airbnb Open Data file `AB_NYC_2019.csv`.

The dataset includes information such as:

- Listing ID and host details
- Borough and neighborhood
- Latitude and longitude
- Room type
- Price
- Minimum nights
- Number of reviews
- Reviews per month
- Availability throughout the year

After cleaning, the final dataset contained **48,392 listings and 20 columns**.

3 Data Cleaning

The raw dataset was cleaned in `01_data_cleaning.ipynb`.

Main cleaning steps included:

- Selected relevant columns for analysis and modeling
- Filled missing values in `name`, `host_name`, `last_review`, and `reviews_per_month`
- Removed listings with price equal to 0
- Removed extreme price outliers above the 99th percentile
- Created engineered features such as:
 - `is_private_room`
 - `is_entire_home`
 - `has_reviews`
 - `price_per_review`

The cleaned dataset was saved as `airbnb_cleaned.csv`.

4 Exploratory Data Analysis

Exploratory analysis was completed in `02_exploratory_data_analysis.ipynb`.

Key findings:

- Entire home/apartment listings had the highest average price.
- Manhattan had the highest average Airbnb price among all boroughs.
- Brooklyn had a high number of listings and more affordable prices compared to Manhattan.
- The price distribution was right-skewed, meaning most listings were in the lower and mid-range price categories, while a smaller number of luxury listings increased the average price.

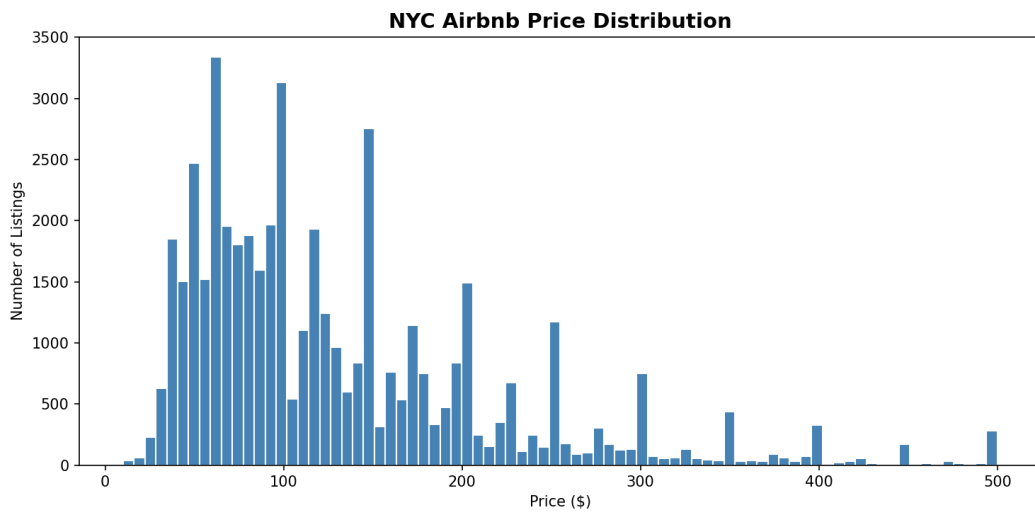


Figure 1: NYC Airbnb Price Distribution

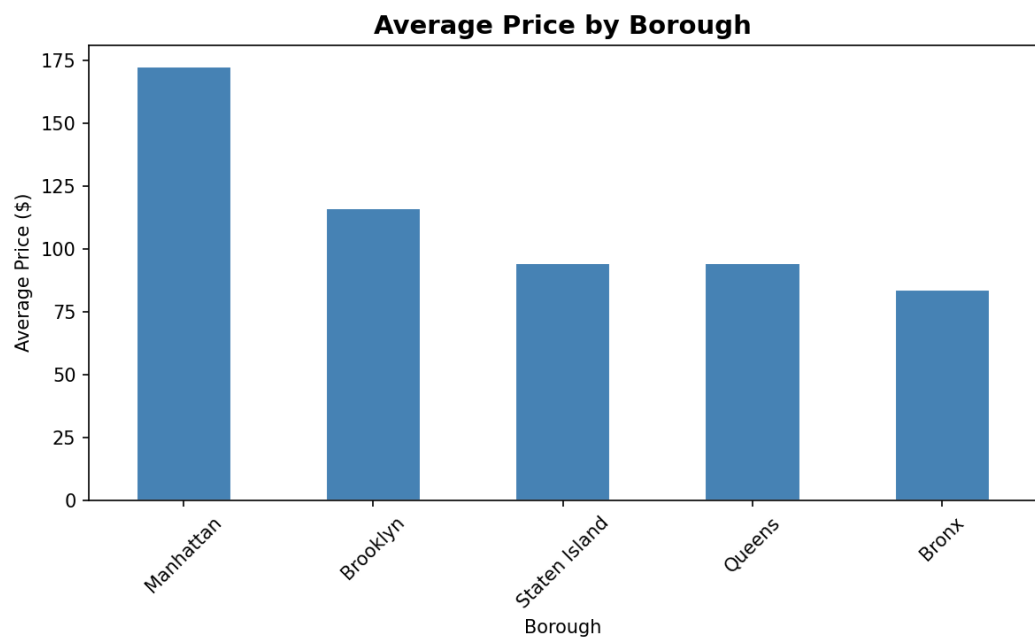


Figure 2: Average Airbnb Price by Borough

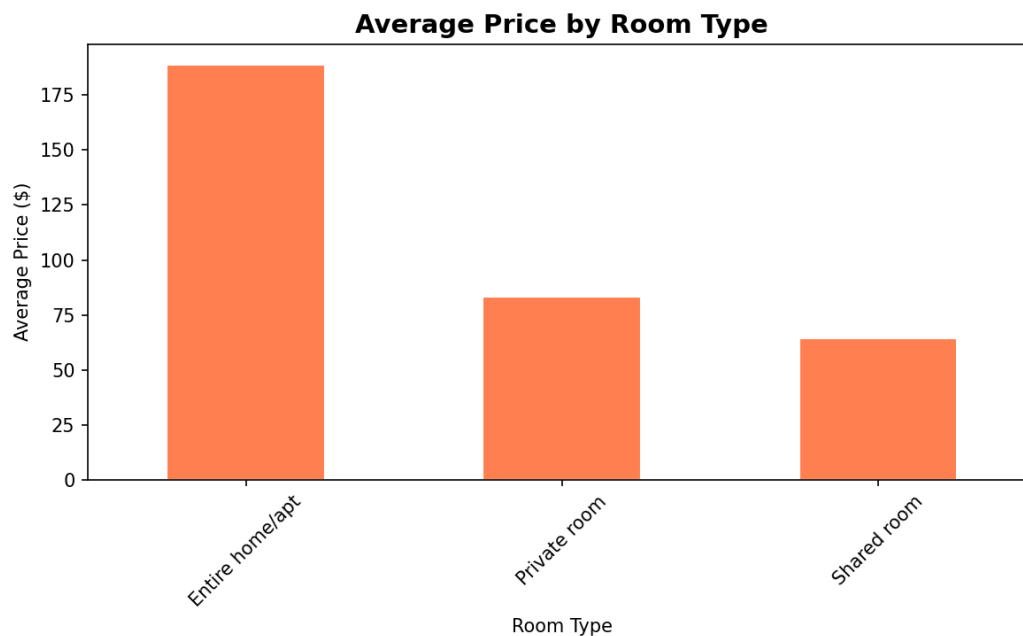


Figure 3: Average Airbnb Price by Room Type

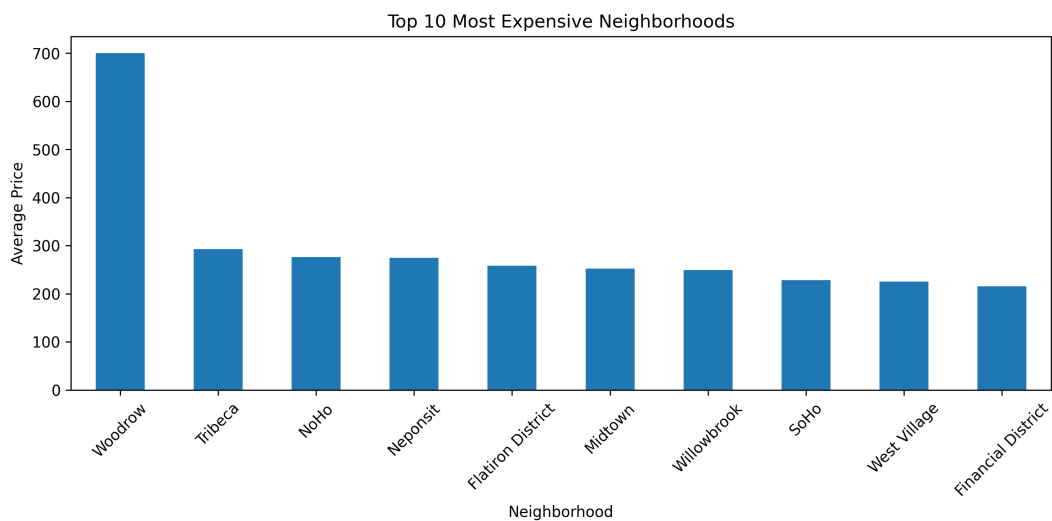


Figure 4: Top 10 Most Expensive NYC Neighborhoods

5 Machine Learning Modeling

Machine learning was completed in `03_modeling.ipynb`.

Two models were trained:

- Linear Regression

- Random Forest Regressor

The target variable was `price`.

Features used included:

- `neighbourhood_group`
- `neighbourhood`
- `room_type`
- `minimum_nights`
- `number_of_reviews`
- `reviews_per_month`
- `calculated_host_listings_count`
- `availability_365`
- `is_private_room`
- `is_entire_home`
- `has_reviews`

The best model was Random Forest.

Model Result

- Mean Absolute Error: **\$47.08**
- R^2 Score: **0.46**

The model was also used to identify potentially underpriced listings by comparing actual prices with predicted prices.

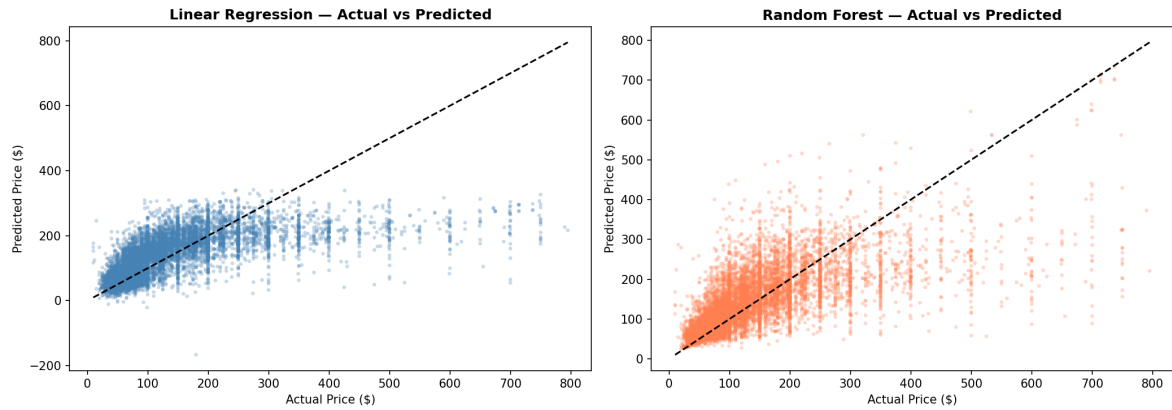


Figure 5: Actual vs Predicted Airbnb Prices

6 Tableau Dashboard

A Tableau dashboard was created to present the business insights visually.

The dashboard includes:

- Listings by borough
- Average price by room type
- Average price by borough
- NYC Airbnb listing map

The dashboard helps users quickly understand pricing patterns and listing distribution across New York City.

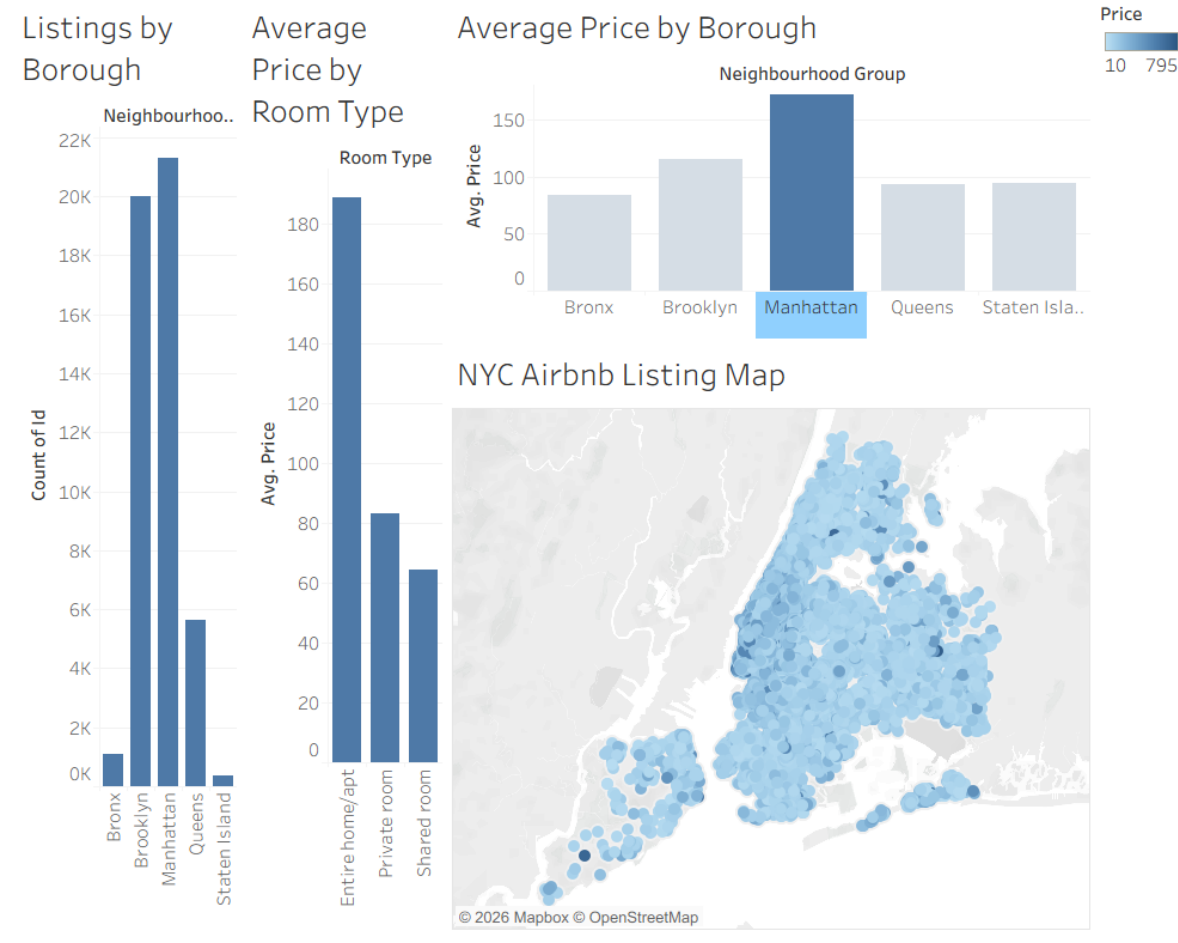


Figure 6: Tableau Dashboard Preview

7 Challenges and Learnings

One important challenge was identifying potential data leakage.

The feature `price_per_review` was created using the `price` column. Since `price` is the target variable, using `price_per_review` in the machine learning model would give the model information from the target and artificially improve performance.

To avoid this, `price_per_review` was excluded from model training. It was kept only for exploratory analysis and dashboard insights.

This helped make the model evaluation more realistic and reliable.

8 Business Insights

The main business insights from the project are:

- Location is one of the strongest factors affecting Airbnb prices.
- Entire homes and apartments are significantly more expensive than private rooms and shared rooms.
- Manhattan listings have a strong price premium.
- Brooklyn is a strong high-volume market with more competitive prices.
- Machine learning can help estimate fair prices and identify potentially underpriced listings.

9 Conclusion

This project successfully completed the full data science workflow: data cleaning, exploratory analysis, machine learning modeling, dashboard creation, and business recommendation development.

The final output can help Airbnb hosts understand market pricing, compare neighborhoods, and make better pricing decisions.